# Day 1 Deliverable: Methodology Review and Prototype Completion

Insider Threat Detection System, Capstone Group 7 · 5 November 2026

## 1. Purpose

This document is the Day 1 deliverable. It answers both columns of the task table.

| Column | Day 1 task | Where answered |
|---|---|---|
| Research and methodology | Review methodology for missing technical details and implementation dependencies | Sections 2 and 3 |
| Prototype development | Complete any remaining essential frontend and backend components | Sections 4 and 5 |

Evidence (test results, dependency lock file, Python version, code check) is in Section 6.

## 2. Methodology review: missing technical details

The review compared the written methodology (METHODOLOGY.md, DATA_RULES.md, TOOLS_AND_ALGORITHMS.md, README.md and the Word report) with the code in backend/itd (ingest, sessionize, features, baselines, models, risk, engine, api, schema). Each gap below was found by reading both sides and checking that they agree.

| # | Gap found | Why it matters | Action | Status |
|---|---|---|---|---|
| M1 | Python dependencies have lower bounds only (for example scikit-learn>=1.4) and there is no lock file | A rebuild can install other versions and give other scores | Save the resolved versions in backend/requirements.lock.txt | Done (Section 6, Figure 1) |

| # | Gap found | Why it matters | Action | Status |
|---|---|---|---|---|
| M2 | Python and MongoDB versions (first listed as a gap by mistake) | A reader cannot reproduce the environment | Checked: TOOLS_AND_ALGORITHMS.md already states Python 3.14.6 and the image mongo:7. A second environment (Python 3.14.5, Docker 29.9.2) was added there. | Closed (listed by mistake) |
| M3 | A missing column or a wrong date format stops the run with a raw pandas error; duplicate event ids and non-numeric email sizes are accepted silently | Bad input is not reported clearly and can change results unnoticed | Pre-flight validation with a saved report | Done on Day 2 (itd/validate.py) |
| M4 | The engine can only be started from the command line; the dashboard cannot start or monitor a run | The core processing module is not connected to the interface | Run endpoint, run status and a run button | Done on Day 3 (run request, worker, Run analysis panel) |
| M5 | Original CERT r4.2 and SPEDIA have not been run; every number comes from generated data | Results must not be presented as research findings | State this limit in every result | Open, stated |

| # | Gap found | Why it matters | Action | Status |
|---|---|---|---|---|
| M6 | SHAP explains the XGBoost model only, not the combined score | A reader could think the explanation covers all three models | Keep the sentence "What drove the XGBoost score" and state it in the methodology | Already in the dashboard |
| M7 | The time-ordered train, validation and test split by position is correct only if sessions are sorted by start time | An unsorted list would leak future sessions into training | Confirmed: sessionize.py sorts by start; state it as an assumption | Checked |

# 3. Implementation dependencies

## 3.1 Software dependencies

Versions below are the minimum versions in the requirement files. The versions actually installed for this run are saved in backend/requirements.lock.txt.

| Layer | Dependency | Constraint in repository |
|---|---|---|
| Language | Python | 3.14.5 used in this run |
| Data | pandas, numpy | >=2.2, >=1.26 |
| Models | scikit-learn, xgboost, shap, joblib | >=1.4, >=2.0, >=0.45, >=1.3 |
| Backend | fastapi, uvicorn, httpx | >=0.110, >=0.29, >=0.27 |
| Database | pymongo, MongoDB server (Docker) | pymongo >=4.6 |
| Security | bcrypt, PyJWT | >=4.1, >=2.8 |
| Tests | pytest, mongomock, cryptography | >=8.0, >=4.1, >=42.0 |
| Frontend | react, react-dom, vite, @vitejs/plugin-react | ^18.3.1, ^18.3.1, ^5.4.11, ^4.3.4 |

### 3.2 Run-order dependencies

These dependencies are not obvious from the library list. They were read from the code.

| Step | Needs | What happens if it is missing |
|---|---|---|
| Engine run | MongoDB running; the five CSV files logon, device, email, file, http | The run stops with "Missing in <folder>" or a connection error |
| Training | At least one malicious session in each of train, validation and test | The run stops: "Each of train/validation/test needs at least one malicious session" |
| Labels | answers.csv (optional) | Without it every label is 0 and training is not possible |
| Validation rules in the database | python -m itd.schema apply | Documents are stored without database-level checks |
| API results | A finished engine run | GET /api/run/latest returns 404 "No batch run recorded yet" |
| Login | An analyst account in analyst_accounts | Login returns 401 |
| Assistant (hosted mode) | LLM_API_KEY | The offline template answer is used |

## 4. Prototype: remaining components

The repository was checked component by component against the table task "complete any remaining essential frontend and backend components".

| Component | State at Day 1 | Day 1 result |
|---|---|---|
| Ingestion, sessions, 39 features, baselines | Complete | No change |
| Models, calibration, risk score, SHAP | Complete | No change |
| REST API (login, sessions, user history, latest run, assistant, health) | Complete | No change |
| Dashboard (login, flagged list with CSV download, session detail, user page, run panel) | Complete | No change |

| Component | State at Day 1 | Day 1 result |
|---|---|---|
| Database schema, indexes, run links | Complete | Allowed value "scored" added (P2) |
| Engine result storage | Gap found | Fixed (P1, P3) |
| Input validation | Gap | Day 2 |
| Starting a run from the dashboard | Gap | Day 3 |

### 4.1 Changes made on Day 1

| # | Problem | Fix | File |
|---|---|---|---|
| P1 | The engine deleted all results before writing new ones, so a crash left an empty dashboard | Results with the same session ids are replaced block by block, and results of older runs are deleted only after all new ones are saved | engine.py |
| P2 | Without --retrain, sessions were labelled train, validation or test although the saved model never trained on them | They are labelled "scored"; the database rule accepts the new value | engine.py, schema.py |
| P3 | The number 39 was typed into a progress message | The message uses len(FEATURES) | engine.py |

A first version of P1 used a bulk replace operation. The automated tests showed that the test database does not support it (22 tests failed). It was replaced by the delete and insert version above, which works with both the test database and a real MongoDB.

Known limit of P1: if a run stops in the middle, the dashboard shows a mix of old and new results until the next complete run. Day 3 adds a run status so the dashboard can show this.

## 5. Test history

| Run | Result | Meaning |
|---|---|---|
| After the first version of P1 | 20 passed, 1 skipped, 22 errors | The bulk replace operation is not supported by the test database |

| Run | Result | Meaning |
|---|---|---|
| After the corrected P1 | 42 passed, 1 skipped, 1 warning in 12.81 s | All earlier behavior is kept; the skipped test needs a real MongoDB |

## 6. Evidence

Figure 1 shows three results in one terminal session: (a) the code check of engine.py, where line 130 replaces results with the same ids, line 131 inserts the new results and line 132 deletes results of other runs; (b) the test run with 42 passed and 1 skipped; (c) the saved lock file and the Python version 3.14.5.

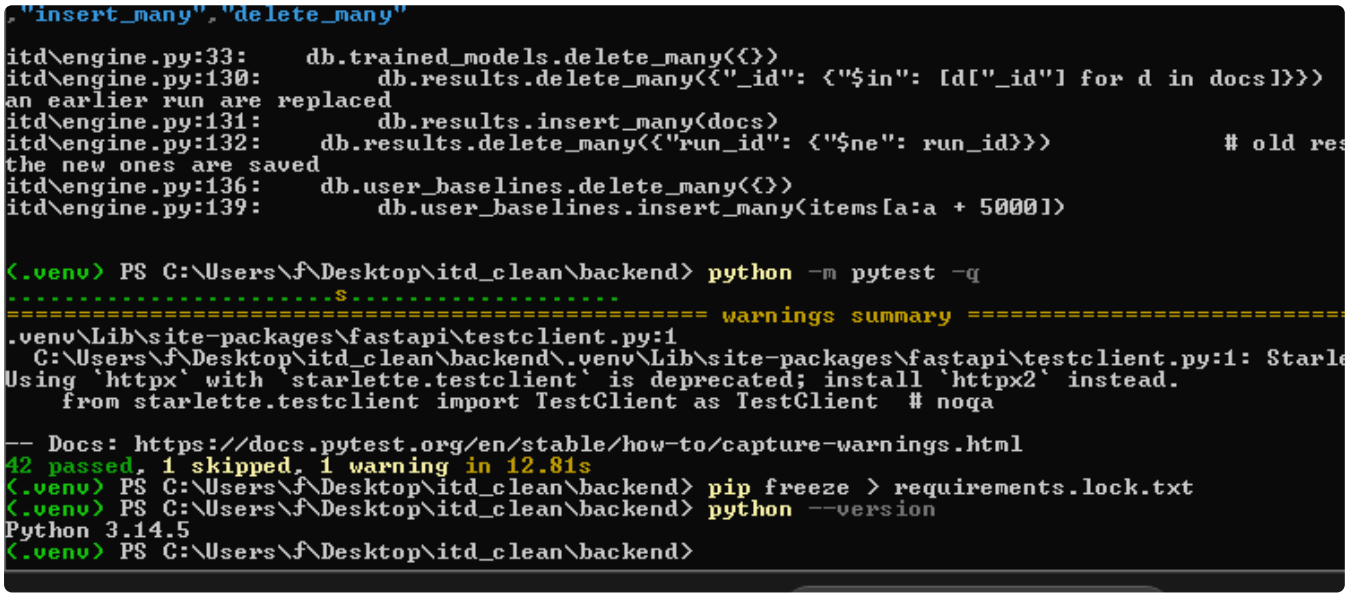



Figure 1. Code check, test result, lock file and Python version.

Figure 2 shows the creation of the review document DAY1_REVIEW.md in the docs folder of the repository.

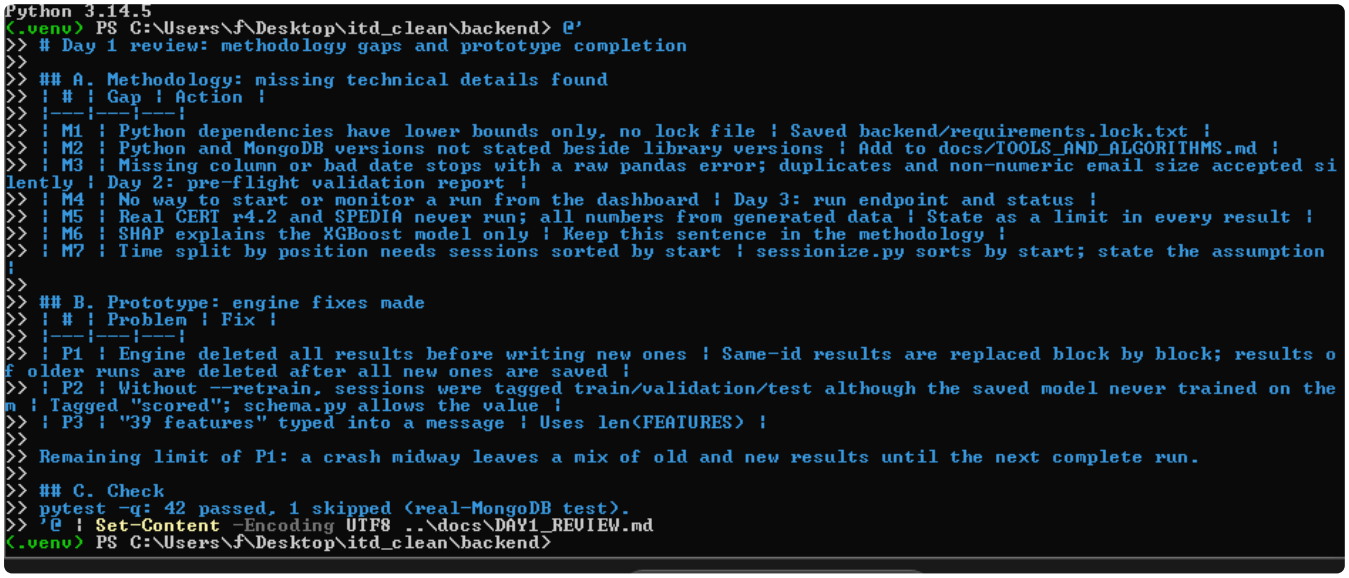


Figure 2. Creation of the review document DAY1_REVIEW.md.

## 7. Open items and later verification

- Closed on 6 November 2026 against a real MongoDB: after python -m itd.schema apply, a run started from the dashboard stored 7,329 sessions that all carry the label scored, because the run reused the saved models.
- The test that needs a real MongoDB is skipped by default.
- All data is generated CERT-layout data. Original CERT r4.2 and SPEDIA have not been run.

## 8. Next steps

| Day | Research and methodology | Prototype development |
|---|---|---|
| 2 | Dataset sources, collection procedures and input-data requirements | Data import, preprocessing and validation mechanisms |
| 3 | Data-processing steps, algorithm configuration and output generation | Connect the main processing module to the system interface |